The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, SPRING SEMESTER 2022-2023

COMPUTER SECURITY
(COMP3028)

Time allowed ONE Hour

Candidates must NOT start writing their answers until told to do so

Answer ALL THREE questions

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

INFORMATION FOR INVIGILATORS: Please ensure that the Exam Paper is also collected at the end of the examination.

1. The following questions are about computer security in general.

a) You are the IT manager of a company that provides laptop PCs to its sales employees. You are concerned about the security implications. This is because the sales staff can store sensitive data on their laptop PCs and then use them for email.

Identify TWO (2) risks to data on a laptop PC and briefly explain how each risk can compromise the confidentiality, integrity or availability of the data.

(4 Marks)

b) Biometrics is used in computer and network security.

i) Briefly explain what is meant by biometrics in relation to authentication.

(3 marks)

ii) State TWO (2) types of biometric with ONE(1) example for each type.

(4 Marks)

c) Cryptanalysis is the art or process of deciphering coded messages without being told the key. A form of Cryptanalysis is the Brute Force Attack.

i) Briefly explain how a Brute Force Attack works.

(2 marks)

ii) With the use of an example, outline the effect of different key sizes in relation to a Brute Force Attack.

(3 Marks)

d) Explain how a SYN spoofing attack works and its effect.

(4 marks)

2. The following questions are about public-key cryptosystem, hash function, and modes of operation.

RSA is a well-known public-key cryptosystem. An RSA system uses two large primes, *p* and *q*, and a cryptographic hash function H(x) (e.g. SHA-1). Answer the following questions:

a) Explain how the RSA public and private keys are generated, and give the necessary equations.

(4 marks)

b) Give the equation to encrypt a message M (assuming M is shorter than the product *pq*) to achieve confidentiality using the RSA cryptosystem, and the equation to decrypt the ciphertext to obtain the plaintext.

(2 marks)

c) Now assume M is larger than the product *pq*. Use block diagrams to illustrate how the message is encrypted and decrypted using the ECB (Electronic Codebook) and CBC (Cipher Block Chaining) modes, respectively.

(6 marks)

d) Give a method to achieve anti-replay and non-repudiation of transmission using the RSA cryptosystem (details such as the processes performed at the sender and at the receiver should be given). Name two factors on which the security of this method depends.

(8 marks)

3. The following questions are about passwords and network security.

a) Several million user names and passwords from ABC.com were leaked by hackers and put up on a web forum. The passwords were hashed but not salted.

i) What can the hackers or anyone that downloaded the files do to get to the plaintext passwords?

(2 marks)

ii) What if the passwords had been salted - with the same salt for each user?

(2 marks)

iii) Give THREE(3) recommendations and reasons to the ABC security team to best protect user passwords.

(6 marks)

b) Explain the strengths and weaknesses of each of the following firewall deployment scenarios in defending servers, desktop computers, and laptops against networks threats.

i) A firewall at the network perimeter (i.e., where the network meets the Internet).

(4 marks)

ii) Firewalls on every end host machine.

(4 marks)

c) Describe what a phishing attack is and how it can break password security.

(2 marks)